

TED(10)-3071

REVISION-2015

Model question

THIRD SEMESTER DIPLOMA EXAMINATION IN CHEMICAL ENGINEERING

CHEMICAL PROCESS PRINCIPLE

Maximum marks: 100

Time : 3hrs

PART-A

(Maximum marks: 10)

Answer all questions. Each question carries 2 marks

- I
- a) State Charles's law
 - b) State the numerical value of "R" the universal gas constant in lit. atm/mol K
 - c) State the material balance equation with accumulation and without accumulation
 - d) Define limiting reagent?
 - e) Define open system
- 5x2=10

PART-B

(Maximum marks: 30)

- II Answer any five of the following question. Each question carries 6 marks

- a) i) compute ideal gas equation using gas laws 3
ii) 15kg of carbon dioxide is compressed at a temperature of 303K (30°C) to a volume of 0.5 m³. Calculate the pressure required for given duty. 3
Assume ideal gas law is applicable
- b) State the following
 - i) Boyle's law 2
 - ii) Amagat's law 2
 - iii) Calculate the density of air containing 21% O₂, 79% N₂ by volume at 503 K (230°C) and 1519.875 kpa 2
- c) Explain evaporation operation with material balance equation and block diagram 6
- d) The NH₃ air mixture containing 0.2kg NH₃/Kg of air enters into absorption system where ammonia is absorbed in water. The gas leaving the system is found to contain 0.004 kg NH₃/Kg of air. Find percentage recovery of ammonia 6
- e) In the production of sulphur trioxide, 100 kmol of SO₂ and 200 kmol of O₂ are fed to reactor. The product stream is found to contain 80 6

kmolSO₃. Find percent conversion of SO₂ and excess amount of O₂ ?

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| f) | Define the following | |
| | i) Stoichiometric ratio | 2 |
| | ii) Excess reagent | 2 |
| | iii) % yield | 2 |
| g) | State the following | |
| | i) Heat of formation | 2 |
| | ii) Heat of Combustion | 2 |
| | iii) entropy | 2 |

PART-C

(Maximum marks: 60)

(Answer one full question from each unit. Each question carries 15 marks)

Module-1

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| III | a) | A gas mixture has the following composition by volume.
Ethylene:30.6%,Benzene: 24.5%,Oxygen: 1.3%,Methane:15.5%,
Ethane: 25%, Nitrogen: 3.1%. Find (a) the average molar mass of the
gas mixture (b) the composition by mass (c)the density of the mixture
in Kg/m ³ at NTP | 8 |
| | b) | A sample of a gas having volume of 0.5m ³ is compressed in such a
manner so that the pressure is increased by 60%. The operation is
done for a fixed mass of a gas at constant temperature. Calculate the
final volume of the gas | 7 |
| | | Or | |
| IV | a) | In the manufacture of nitric acid initially ammonia and air are mixed at 7 atm
and 650°C. The composition of the mixture (on volume basis) is as follows.
Nitrogen=70.5%, Oxygen=18.8%, Water=1.2%, Ammonia=9.5%. Find the
density of gas mixture using ideal gas law | 8 |
| | b) | A gas mixture has following composition by volume. O ₂ : 15.1% CO ₂ : 3.7%
N ₂ : 75%, and H ₂ O: 6.2%. Find | 7 |
| | i) | Average molecular weight of the gas mixture? | |
| | ii) | Composition by weight? | |
| | iii) | Density of gas mixture at 100.658kpa and 303 K? | |

Module-II

- V a) The feed containing 50% Benzene and 50% Toluene is fed to a distillation column at the rate of 5000kg/h. A top product contains 95% Benzene and the bottom product contains 92% toluene by weight. Calculate (a) the mass flow rate of top and bottom product (b)% recovery of benzene 8
- b) The waste acid from a nitrating process contains 25% HNO_3 , 50% H_2SO_4 , and 25% H_2O by weight. The acid is to be concentrated to contain 30% HNO_3 and 60% H_2SO_4 by the addition of concentrated 95% H_2SO_4 and concentrated 90% HNO_3 . Calculate the weight of waste acid and concentrated acid which must be mixed to obtain 10000kg of desired composition 7
- Or
- VI a) 2000kg of wet solids containing 70% solids by weight are fed to a tray dryer where it is dried by hot air. The product finally obtained is found to contain 1% moisture by weight. Calculate (a) kg of water removed from wet solids (b) Kg of product obtained 7
- b) A soyabean seeds are extracted with Hexane in batch extractors. The flaked seed contain 18.6% oil, 69.6% solid and 12.4% moisture. At the end of the process, cake separated from hexane oil mixture. The cake analysis yield 0.8% oil, 87.7% solids and 11.5% moisture. Find the % recovery of oil. All% are by weight 8

Module-III

- VII a) A coal has ultimate composition C 67.34%, H_2 4.67%, O_2 8.47%, N_2 1.25%, and S 4.77% and the rest is ash. Find the theoretical air-fuel ratio? 8
- b) In production of SO_3 and 100kmol SO_2 and 100kmol O_2 are fed to the reactor. If the % conversion of SO_2 is 80. Calculate the composition of product stream on mol basis 7
- Or
- VIII a) The reaction between ethylene and hydrogen bromide is carried out in a continuous reactor. The product stream is analyzed and found to contain 50mol% $\text{C}_2\text{H}_5\text{Br}$ and 33% HBr . The feed to the reactor contain only ethylene and hydrogen bromide. Calculate the fractional conversion of the limiting reactant and the % by which the other reactant in excess 7
- b) A furnace is fired with fuel oil. The Orsat analysis of the flue gas 8

indicated 10.6%CO₂,6% O₂and rest N₂ by volume. Find C: H ratio in the fuel oil. assuming that fuel oil does not contain Nitrogen

Module- IV

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| IX | a) State the following | |
| | i) State function | 2 |
| | ii) Internal energy | 2 |
| | iii) Second law of thermo dynamics | 2 |
| | iv) Hess's law | 2 |
| | b) Calculate the heat of formation of phenol crystals at298.15K(25 ⁰ c)
from the elements using following data
standard heat of formation of CO ₂ (g) = -393.5kJ/mol
standard heat of formation of H ₂ O(l) = -393.5kJ/mol
heat of combustion of Phenol Crystals at 298.18K(25 ⁰ c) =
-3053.51kJ/mol | 7 |
| X | a) Define the following | |
| | i) Heat capacity | 2 |
| | ii) Extensive property | 2 |
| | iii) Closed system | 2 |
| | iv) Third law of thermo dynamics | 2 |
| | b) Calculate the heat of formation of Benzene, given the heat of
combustion of benzene, carbon and hydrogen are-780kcal,-94.0kcal
and -68.0kcalat300K(27 ⁰ c)
Given data: | 7 |
| | i) C ₆ H ₆ (l) +7.5O ₂ (g)→6CO ₂ (g) + 3H ₂ O(l) ΔH =-780kcal | |
| | ii) C(s) + O ₂ (g) → CO ₂ (g) ΔH = - 94.0Kcal | |
| | iii) H ₂ (g) + 0.5 O ₂ (g) → H ₂ O(l) ΔH = -68kcal | |